Optimizing Household Energy Storage and Utility Coordination Under India's PM Surya Ghar Scheme

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Note: Please the excel and python models via the following link:
https://drive.google.com/drive/folders/1s-XiZnq6d3P_qMK8mfJBsKOuVJnbmMLO?usp=drive_link

Problem Background & Motivation

Once upon a time, in some parts of rural India, electricity was a luxury. But today, due to rapid "rural electrification" under the Modi government, 99.5% of the population has electricity. This gives the country the status of "near universal electrification". Clearly, India has made great strides in this regard. However, it didn't fully eliminate them from every problem. Grid reliability still remains a problem due to supply chain shortages & the aging infrastructure. ("Understand Energy in India | Understand Energy Learning Hub") Particularly, states like Rajasthan & Gujarat experience load shedding during peak hours. On the flip side, DISCOMs are struggling with feeder-level demand management. This motivated the Indian Government to launch PM Surya Ghar: Muft Bijli Yojana aiming to achieve rooftop solar installations in over one crore households by FY 2026-27 ("PM Surya Ghar: Muft Bijli Yojana"). However, this new addition brings with a new kind of operational risk.

Imagine what would happen if all the houses charge their batteries at the same time (during midday hours). This is a secondary peak that the grid could face, stressing the infrastructure again rather than relieving it. This behaviour of the households can be backed by the economic theory "Tragedy of the commons" as well. Every individual (the households) act in his/her best interest, that the common resource ends up getting overused. This served as our primary motivation for building an optimization problem around this scenario.

Through this project, we wish to build a demand response mechanism model that could be adopted by PM Surya Ghar households in Gujarat. We consider a hypothetical world with realistic assumptions. Currently, there is no record of a "DISCOM administered battery discharge incentive program" in India. For the purpose of this project, we considered a voluntary-incentive based program coordinated by a local DISCOM, wherein households receive a per kWh payment for discharging their batteries between the peak hours (18:00 to 22:00). This introduces two stakeholder optimization problems. On one side households must decide when to charge, discharge the battery, import from the grid and export surplus PV over the 24 hour time horizon. Ultimately, they need to minimize their total electricity cost. On the other hand, the local DISCOM, as the coordinating unit, must decide on the incentive rate that brings down the feeder peak demand but is also well within the budget. The DISCOM must set the incentive rate such that households respond optimally. Additionally, the incentive budget, as well as the peak reduction outcome depend on how these houses respond. This introduces a bi-level optimization model.

Thus, this study aims to contribute an optimization model that can be applicable to India's evolving landscape. An important point to note is that, all the elements for this hypothetical program (DISCOM administered demand response, per kWh discharge incentives, evening peak windows & household-level

battery optimization) are already individually grounded in real programs & frameworks across the country & the international markets as well.

Model Formulation

For each household i and hour $\in$ {1,.....,24}, the model formulation is as follows:

Level 1: Household Battery Dispatch

- Decision Variables: energy charged (ch), energy discharged (dis), imported from the grid (g), and PV fed back (feed)
- Battery state of charge (SOC) is derived from energy charged (ch) & energy discharged (dis)
- The objective is to minimize the net electricity cost:

$$Min\ (\Sigma_t \lambda_t \cdot g_{i,t} - \lambda_inc \cdot \Sigma_{t \in Peak}\ dis_{i,t} - \lambda_feed \cdot \Sigma_t\ feed_{i,t})$$

Where:

1. λ_t: Time of use of the grid tariff
2. λ_inc: Incentive rate for peak discharge
3. λ_feed: Feed in tariff

- The constraints for this model are:

1. Energy Balance: $PV_t + dis_{i,t} + g_{i,t} = D_{i,t} + ch_{i,t} + feed_{i,t}$
2. SOC Dynamics: $s_{i,t} = s_{i,t-1} + ch_{i,t} - dis_{i,t}$, with $s_{i,0} = SOC_init$
3. SOC Bounds: $SOC_min \cdot E_bat \le s_{i,t} \le E_bat$
4. Charge/Discharge Rate limits: $ch_{i,t} \le P_ch$, $dis_{i,t} \le P_dis$
5. Grid Availability: $g_{i,t}, feed_{i,t} \le G_max \cdot a_t$, where $a_t = 0$ during load shedding
6. Non−negativity: all the decision variables ≥ 0

(Note: The feed_t variable is an important extension to the original plan's formulation. The reason being, without this variable, the balance breaks. The midday solar power generates far more energy than what is actually needed by the households. Thus, feed_t acts as a place for the surplus PV to be sent to (to the grid), keeping the balance intact again.)

Level 2: DISCOM Coordination

We solve this through parametric enumeration as the decision space for λ_inc is one-dimensional and the response is piecewise constant (a step function in λ_inc) too. In Excel, we evaluate 12 discrete values (0, 0.01, 0.2,...., 8 Rs/kWh) and select the best feasible rate accordingly. However, since the response saturates at any positive λ_inc, the precise optimal rate lies somewhere between two consecutive grid

values, which discrete enumeration cannot pinpoint. We therefore use Python to compute the exact optimum via a closed-form formula: λ* = B / saturated peak discharge. This is basically a sensitivity analysis.

The objective of this level will be to maximize the following:

$$\begin{aligned} &Min\ \Sigma_i\ \Sigma_\{t \in Peak\}\ g^{\star}_{i,\square}(\lambda_inc) \\ &s.t.\ \lambda_inc \cdot \Sigma_i\ \Sigma_\{t \in Peak\}\ dis^{\star}_{i,\square}(\lambda_inc) \le B \\ &\lambda_inc \ge 0 \end{aligned}$$

Where:

1. *g*, dis*: household's optimal grid import and peak-hour discharge in response to a given incentive rate λ_inc; B: DISCOM's daily incentive budget*
2.

Input Data and Instance Design

The input data for this model was initially generated with the assistance of a generative AI tool (Claude) to establish a structured starting point. However, all values were carefully reviewed and cross-verified using external sources, including published reports and real-world data ranges. This ensured that the final dataset reflects realistic operating conditions rather than relying solely on AI-generated assumptions.

Specifically, the parameters were validated against several authoritative sources: hourly PV output for a 2 kW rooftop system in Jaipur was retrieved from NREL's PVWatts Calculator (pvwatts.nrel.gov); time-of-use tariffs were drawn from the Rajasthan Electricity Regulatory Commission (RERC) Tariff Order for FY 2023-24; the feed-in rate of 3.26 Rs/kWh follows JVVNL's October 2025 Net Metering Order; load shedding hours were calibrated to PRAYAS ESMI Rajasthan district reports; battery specifications (efficiency, capacity, C-rate) reference IRENA's 2023 Renewable Power Generation Costs report; and household demand ranges follow NITI Aayog's 2021 Rural Electrification Report.

The dataset includes key inputs such as hourly solar generation profiles, household electricity demand, time-of-use grid tariffs, battery specifications, and load shedding schedules. These inputs were selected to capture the main factors influencing household battery usage and utility coordination in a rural energy system.

In terms of implementation, we used both Excel and Python, with different modeling scopes due to tool limitations. In Excel, we modeled a single representative household. This simplification was necessary because Excel Solver has constraints on the number of decision variables it can efficiently handle, making it impractical to directly model multiple households.

To better reflect system-level behavior, we expanded the model in Python to include 10 households. This allowed us to capture interactions across multiple users on the same distribution feeder and provided a more realistic representation of aggregate outcomes.

Results & Insights

The Excel workbook separates parameters, profiles, household dispatch, and DISCOM sweep into four sheets; the Python notebook solves via Gurobi with parametric enumeration.

Lower-level

As an illustrative example, at λ_inc = 1.5 Rs/kWh, a single household achieves an optimal objective of -45.24 Rs/day (net revenue). The optimal schedule charges the battery to full capacity during midday, discharges to meet demand during blackouts at t = 18:00 - 22:00, and exports the remaining surplus PV. In the 10-household instance, each household's peak-hour discharge saturates at 8 kWh (= 4 hours × 2 kWh/h). Per-household feed-in volumes vary markedly, from 11.3 kWh (HH03, low demand) to 7.2 kWh (HH07, farm), confirming that the model captures meaningful heterogeneity.

Upper-level

The DISCOM sweep reveals a sharp discontinuity. At λ_inc = 0, peak discharge is only 42.03 kWh (enough to meet blackout demand). As soon as λ_inc exceeds zero, even at 0.01 Rs/kWh, peak discharge jumps to the physical ceiling of 80 kWh and remains there for all higher values. The reason is that, first, the blackout already forces some peak-hour discharge to meet demand, and second, any positive incentive makes full-depth discharge optimal given no efficiency penalty. Consequently, the DISCOM's optimal λ_inc is not determined by marginal household response but by the budget constraint. At B = 30 Rs/day per HH (Rs 300 total feeder budget), the feasibility boundary falls between λ_inc = 3 and λ_inc = 4. λ^* = 3.75 Rs/kWh delivers 80 kWh of discharge at exactly Rs 240 of budget usage.

This is a policy-relevant finding: once any incentive exists, households saturate their batteries, and λ_inc serves primarily to redistribute rent between the DISCOM and households rather than to induce new operational behavior. The DISCOM should therefore set λ_inc as low as possible while remaining above zero.

Heterogeneity

A naive analysis that assumes 10 identical households predicts total net cost = -452.4 Rs, whereas our heterogeneous 10-HH model yields -438.7 Rs, a 3% overstatement of benefit in the naive case. More strikingly, per-household variance in feed-in (7.2 to 11.3 kWh) is entirely erased by the homogeneity assumption. This supports our decision to model heterogeneity explicitly despite the additional computational cost.

Alignment with Course Materials

This project directly applies several core concepts and techniques from the Advanced Business Analytics course, particularly in optimization modeling and decision analysis. The structure of our model, implementation approach, and analysis all reflect topics covered throughout the course.

Linear Programming (LP)

The foundation of our model is linear programming. Both the household battery dispatch problem and the DISCOM coordination problem are formulated as LPs with continuous decision variables, linear objective functions, and linear constraints. This directly reflects the LP frameworks introduced early in the course, where we learned how to model real-world decision problems using mathematical optimization.

Bilevel / Hierarchical Optimization

Our model extends beyond a standard LP by incorporating a bilevel structure. The upper-level decision (DISCOM setting the incentive rate) influences the lower-level decisions (household battery usage). This reflects real-world hierarchical decision-making and builds on the course's emphasis on modeling complex business environments with multiple stakeholders.

Sensitivity Analysis

We perform a parametric analysis by varying the incentive rate (λ_inc) and observing its impact on outcomes such as feeder peak load and household costs. This is closely aligned with sensitivity analysis techniques taught in class, where changes in key parameters are used to understand trade-offs and system behavior.

Network Flow Interpretation

The energy balance constraint in our model can be interpreted as a flow conservation constraint in a time-expanded network. Energy flows from sources (solar, grid, battery) to demand across time periods, similar to network flow models discussed in class. This connection helps provide additional intuition behind the model structure.

Multi-Objective Trade-offs

Although the model is solved as a single-objective optimization at each level, it captures a clear trade-off between two competing goals: minimizing household electricity costs and minimizing DISCOM peak load. By analyzing different incentive levels, we effectively trace a Pareto frontier, which aligns with the course topic of multi-objective decision-making.

Implementation Using Excel and Python

The project also reflects the course's focus on practical implementation. We use Excel Solver to construct and solve the LP in a structured, transparent format, and Python (Gurobipy) for more flexible and scalable computation. This aligns with the course objective of applying analytical tools to real-world decision problems.

Conclusion

This project successfully built and solved a bilevel optimization model for household solar-battery dispatch and DISCOM incentive design under PM Surya Ghar. Our key finding is that the DISCOM's optimal incentive rate is not driven by marginal household response, since the response saturates at any positive incentive, but instead by the budget ceiling: $\lambda^* = 3.75$ Rs/kWh. We further showed that homogeneity assumptions overstate aggregate benefits by 3% and erase real household-level variance. The framework offers practical guidance for utilities designing demand-response programs in rapidly-electrifying rural markets.